HKCT Higher Diploma in Cybersecurity (Testing and Compliance)

03CIT4067 Cybersecurity Project Bi-weekly Meeting

Minutes of the 2nd Bi-weekly Meeting

Date: 2026-10-20 (Tuesday)
Venue: MUC 507 (face-to-face)
Participants:

Students:
Hui Ka Tung. Troy
Huang Kin Ming, William
Shi Hau Yan, Jessie

Lecturer : Liam Xu

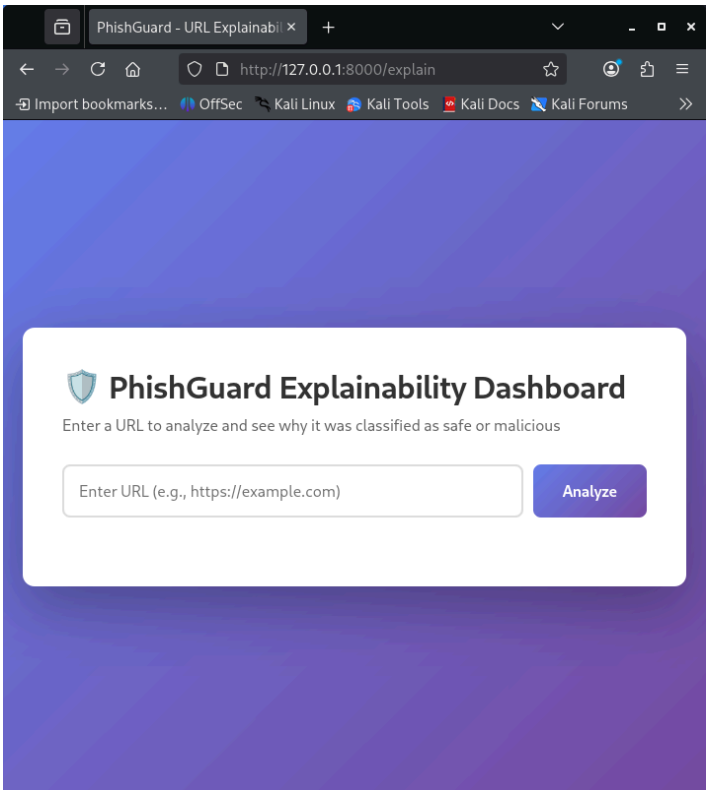
Apologies:

Item	Description	Lecturer Feedback	Action by
Agenda #1	Analysis of Three tools chosen	1. Check each tools accuracy on their own dataset 2. Search for a public phishing dataset 3. Test the public dataset with each tool 4. Compare the accuracy rate of each tool on the public dataset 5. Start to draft the proposal on the	
2.1	Phishing Guard AI		Troy
2.2	PhishingGuard		Jessie
2.3	NLP		William
Agenda #2	Testing Result of those tools		
Agenda #3	Summarize on the strengthens and weakness of those tools		

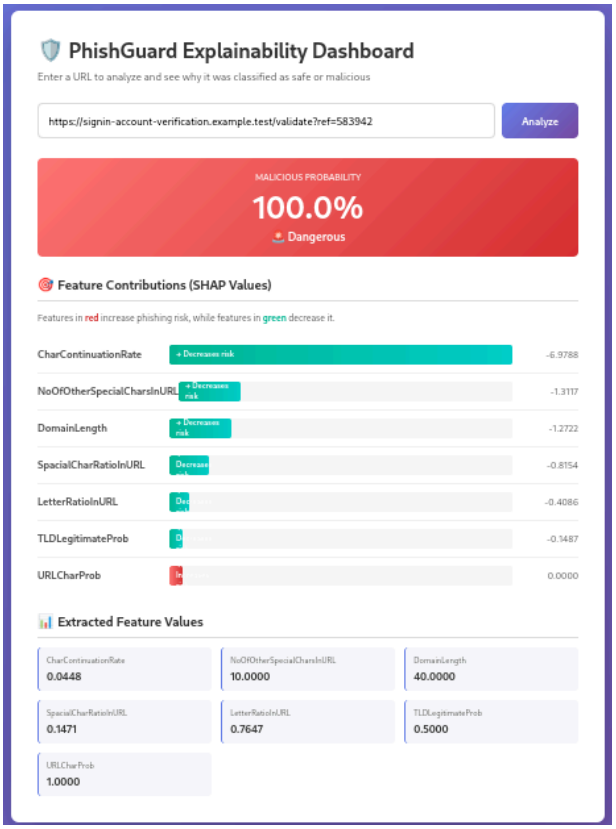
		existing tool analysis, own tool data structure, concepts required to build the own tool	
Agenda #4	Date of next meeing	03/11/2026	

Phishing Guard AI

Interface



- Only allow to upload an URL for testing

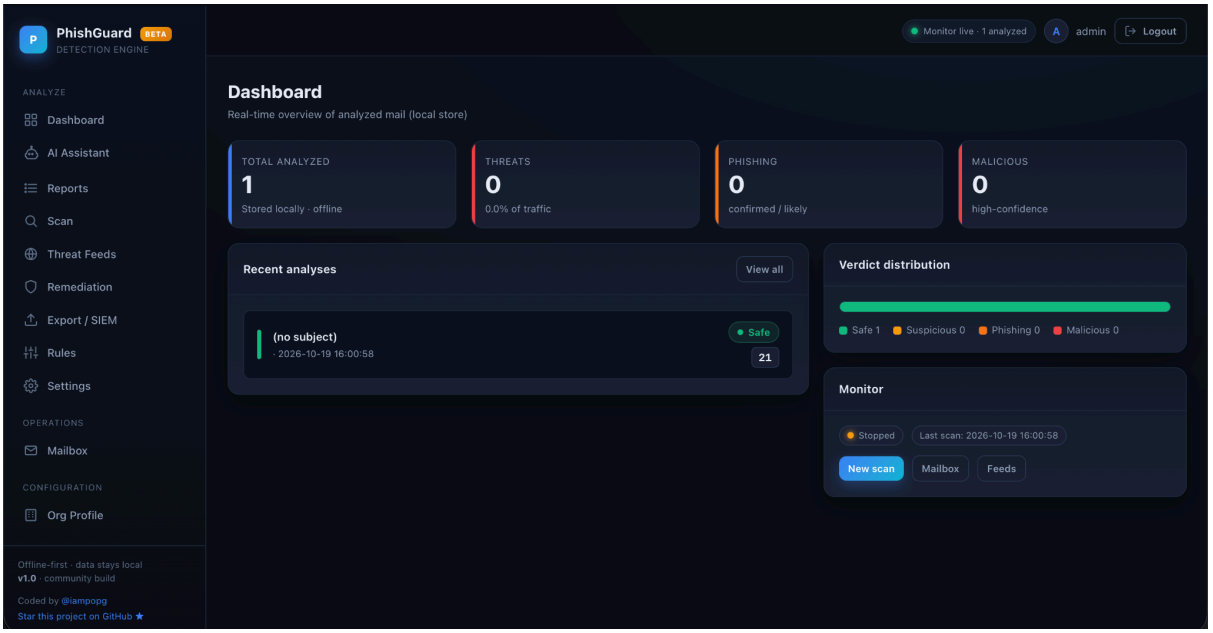


Result Interface

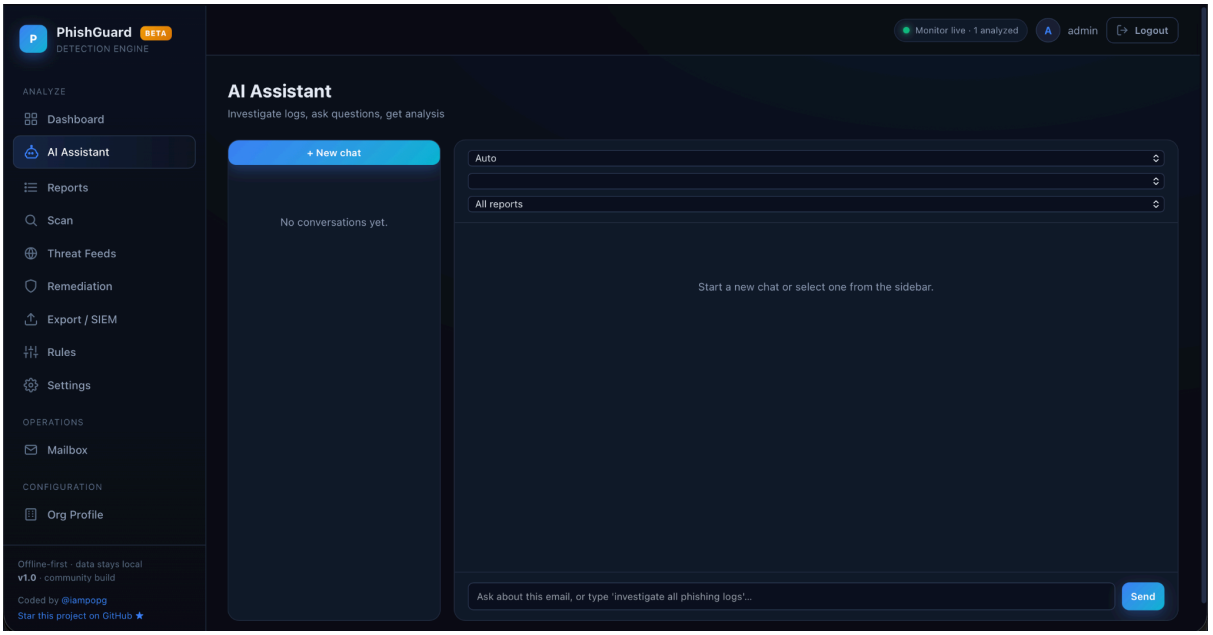
- Green Bar represent decrease the risk
- Red Bar represent increase the risk

PhishGuard

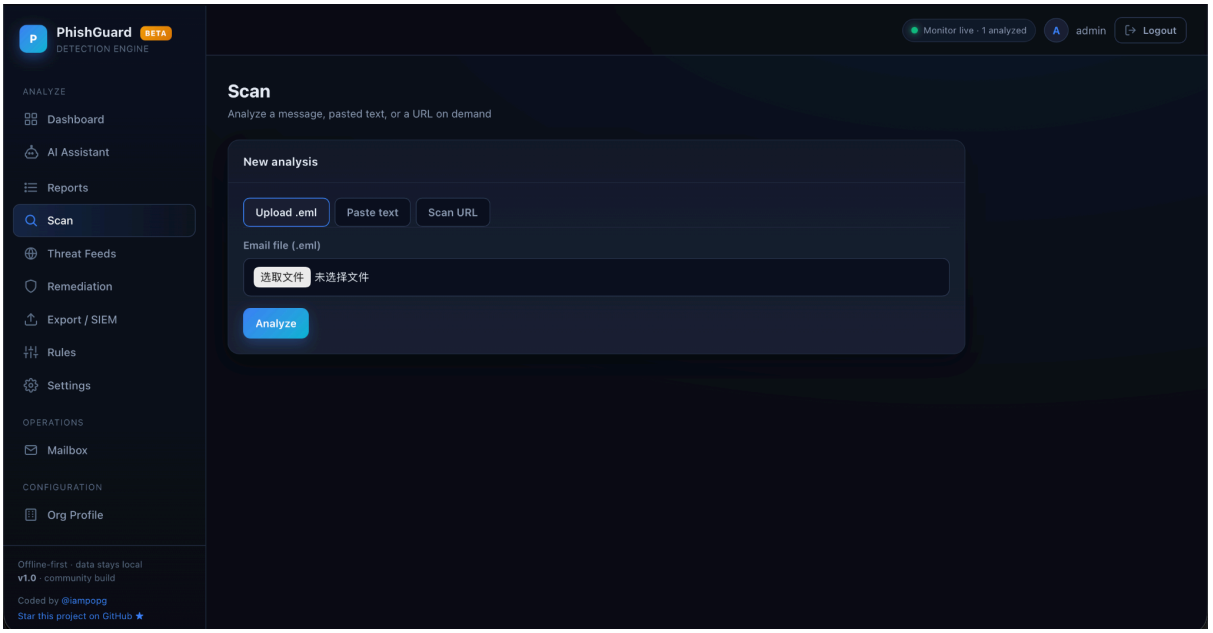
Interface



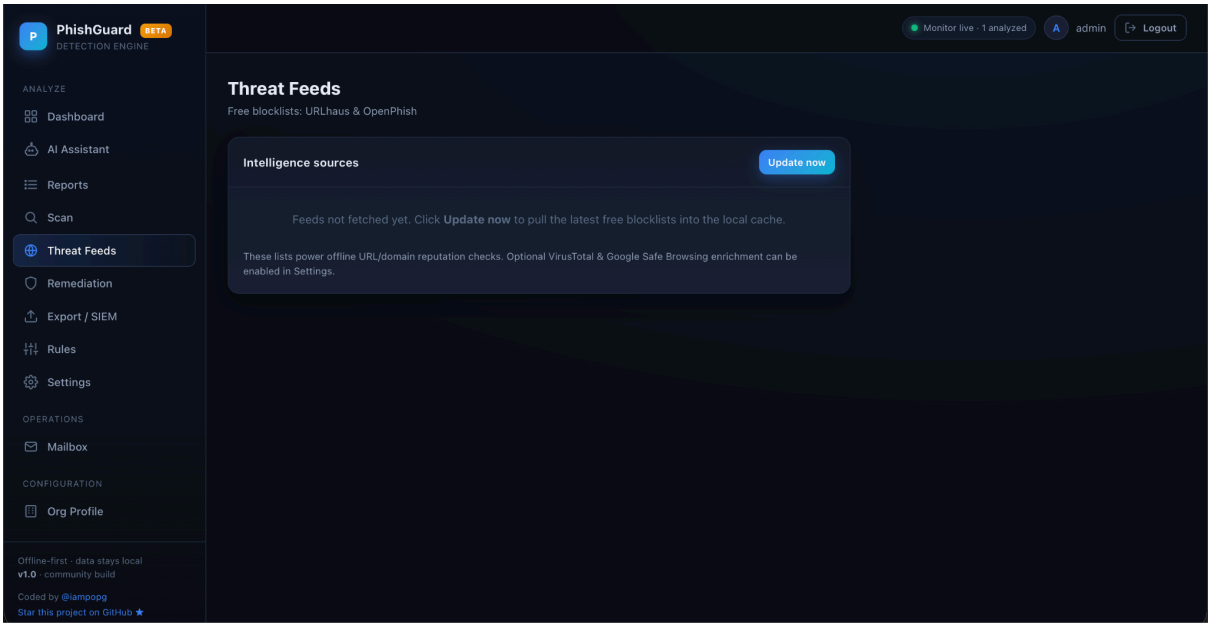
- Dashboard



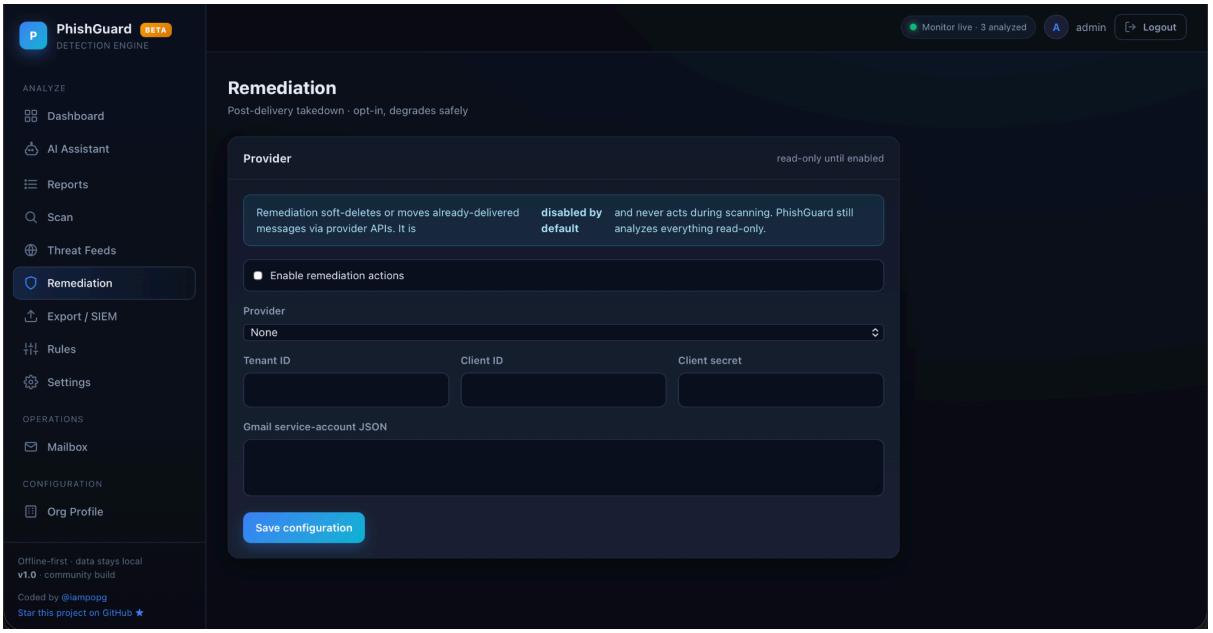
- AI Assistant



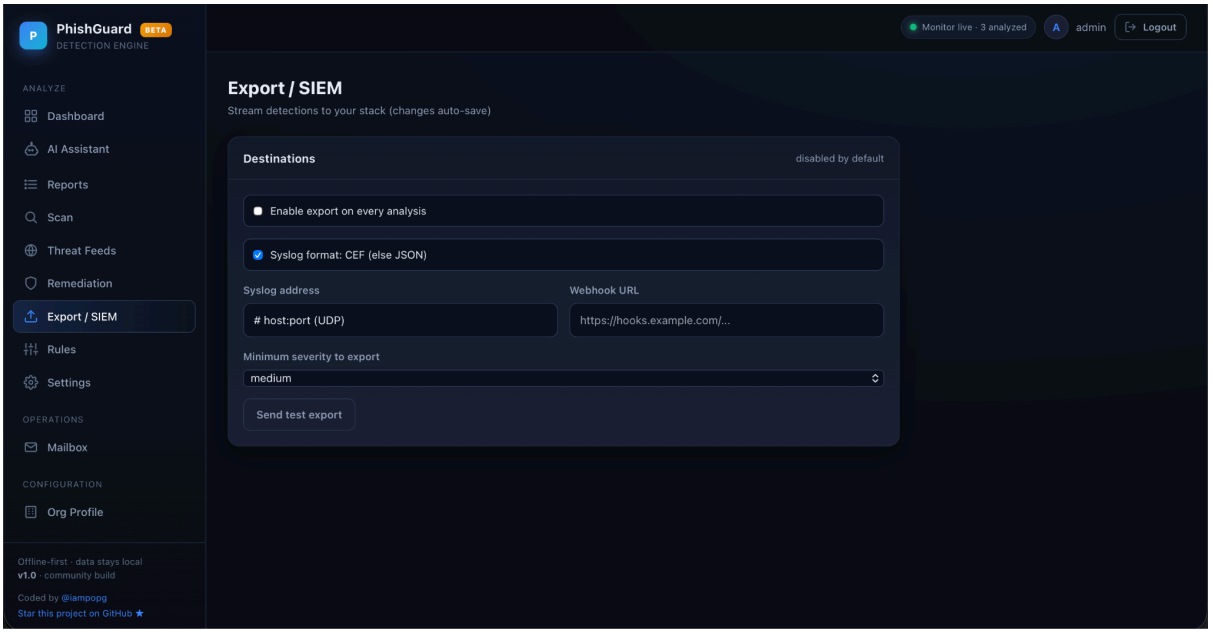
- Scan : by upload the .eml, text, URL or files



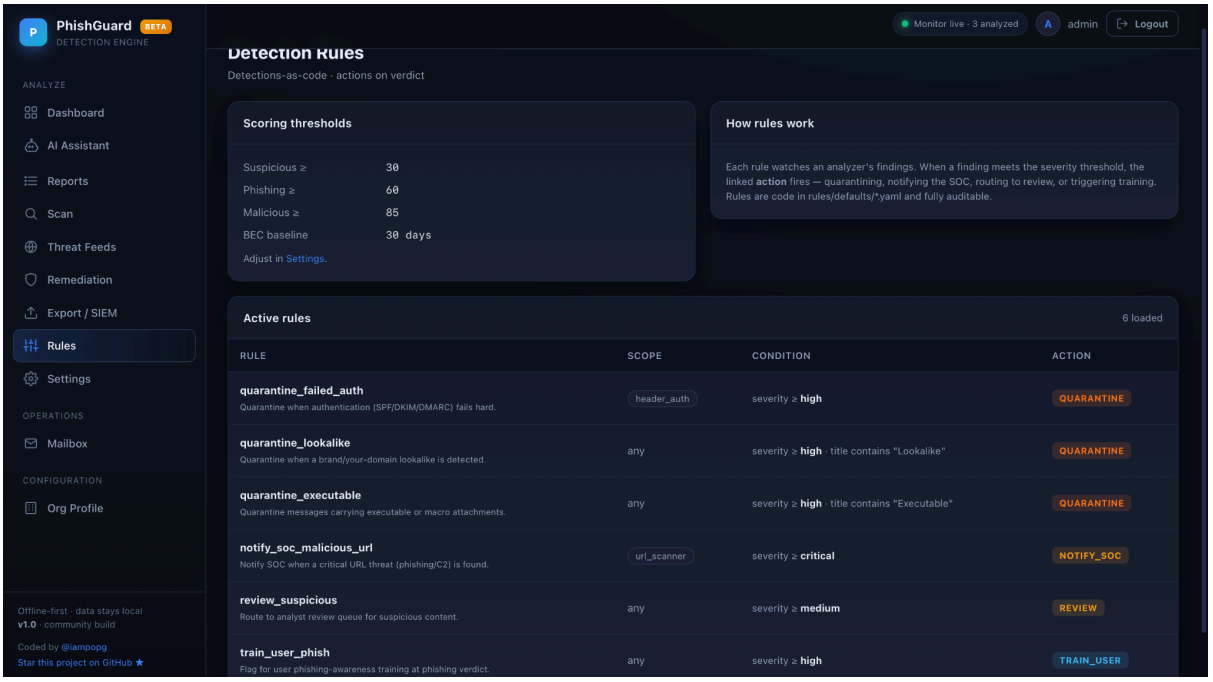
- Threat Feeds : allow user to update the Whitelist



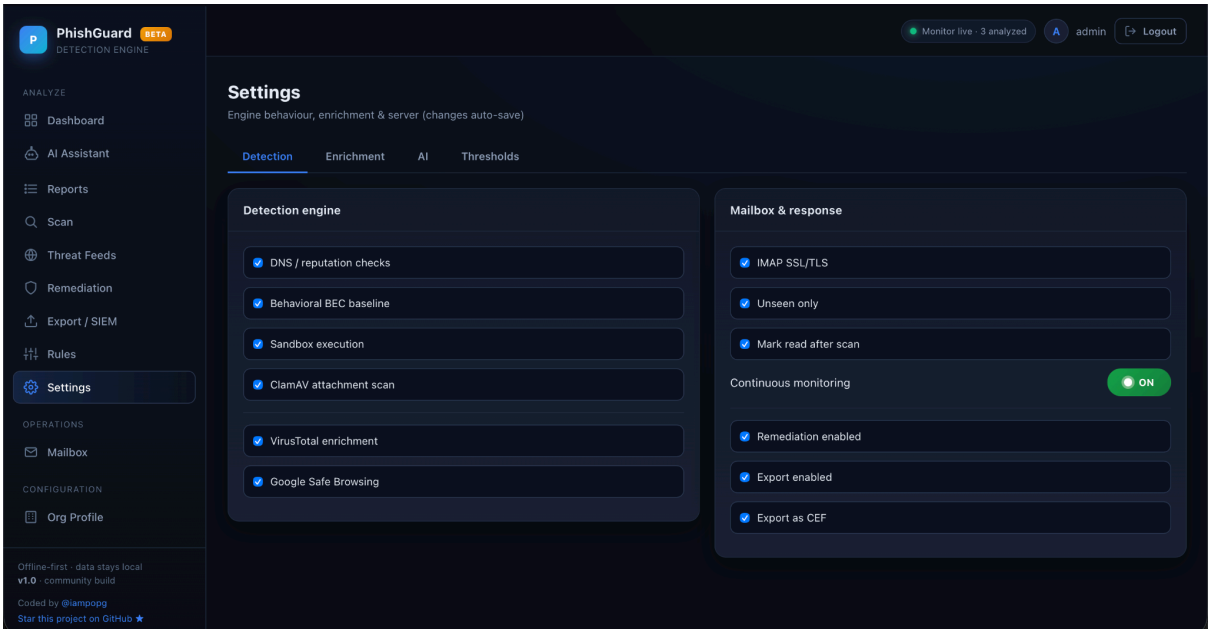
- Remediation 補救/修復



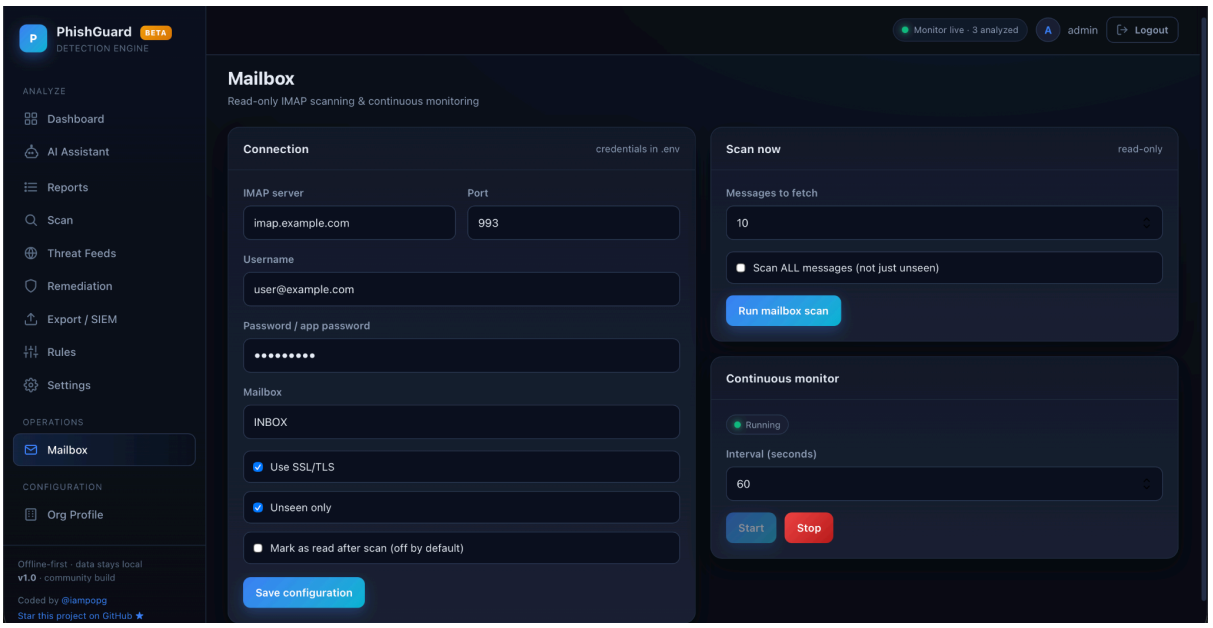
- Export function



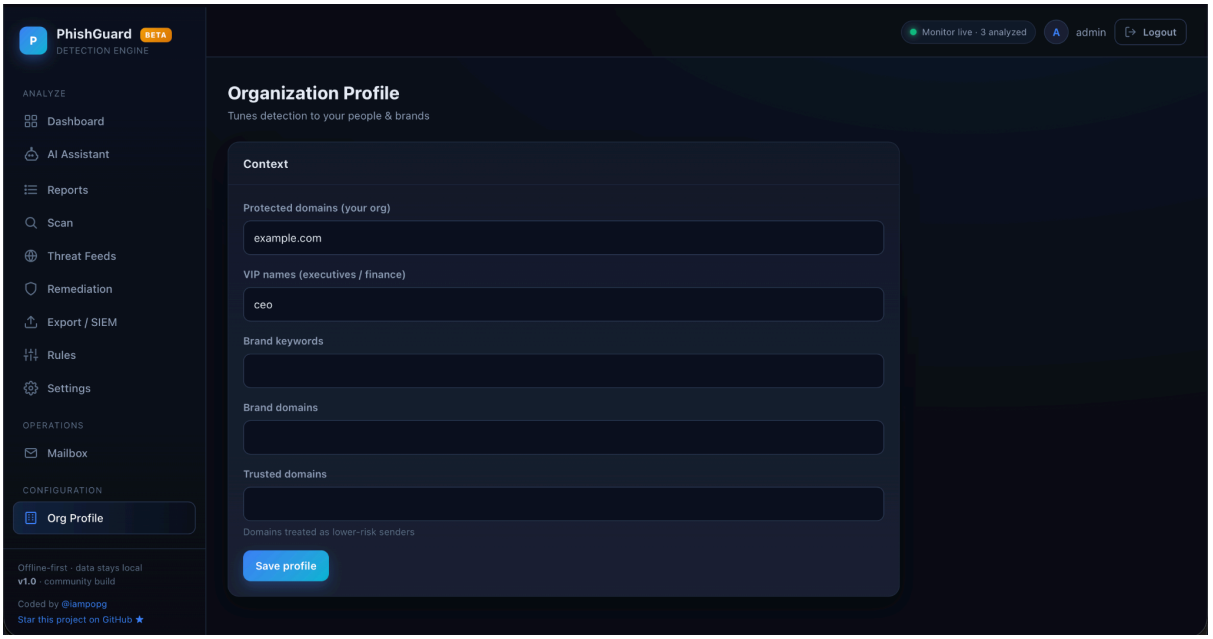
- Rules: allow user to adjust the scoring threshold in setting



- Setting



- Mailbox: Allow user to connect with their mail account



- Org Profile: Allow company to customize the setting

NLP

Interface

[1]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import string
from nltk.corpus import stopwords
import matplotlib.pyplot as plt
```

```
/usr/lib/python3/dist-packages/pyparsing/core.py:5579: SyntaxWarning: 'return' in a 'finally' block
  return f"{type(self).__name__}: {retString}"
```

[2]:
```
df=pd.read_csv("spam.csv",encoding='ISO-8859-1')
```

[3]:
```
df.head()
```

[3]:

	v1	v2	Unnamed: 2	Unnamed: 3	Unnamed: 4
0	ham	Go until jurong point, crazy.. Available only ...	NaN	NaN	NaN
1	ham	Ok lar... Joking wif u oni...	NaN	NaN	NaN
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	NaN	NaN	NaN
3	ham	U dun say so early hor... U c already then say...	NaN	NaN	NaN
4	ham	Nah I don't think he goes to usf, he lives aro...	NaN	NaN	NaN

[4]:
```
df.tail()
```

[4]:

	v1	v2	Unnamed: 2	Unnamed: 3	Unnamed: 4
5567	spam	This is the 2nd time we have tried 2 contact u...	NaN	NaN	NaN
5568	ham	Will ì_ b going to esplanade fr home?	NaN	NaN	NaN
5569	ham	Pity, * was in mood for that. So...any other s...	NaN	NaN	NaN
5570	ham	The guy did some bitching but I acted like i'd...	NaN	NaN	NaN
5571	ham	Rofl. Its true to its name	NaN	NaN	NaN

Right Ctrl

[4]:
```
df.tail()
```

[4]:

	v1	v2	Unnamed: 2	Unnamed: 3	Unnamed: 4
5567	spam	This is the 2nd time we have tried 2 contact u...	NaN	NaN	NaN
5568	ham	Will ì_ b going to esplanade fr home?	NaN	NaN	NaN
5569	ham	Pity, * was in mood for that. So...any other s...	NaN	NaN	NaN
5570	ham	The guy did some bitching but I acted like i'd...	NaN	NaN	NaN
5571	ham	Rofl. Its true to its name	NaN	NaN	NaN

[5]:
```
df.shape
```

[5]:
```
(5572, 5)
```

[6]:
```
df.size
```

[6]:
```
27860
```

[7]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5572 entries, 0 to 5571
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   v1          5572 non-null   object
 1   v2          5572 non-null   object
 2   Unnamed: 2  50 non-null     object
 3   Unnamed: 3  12 non-null     object
 4   Unnamed: 4  6 non-null      object
dtypes: object(5)
memory usage: 217.8+ KB
```

[8]:
```
df.describe()
```

[8]:

	v1	v2	Unnamed: 2	Unnamed: 3	Unnamed: 4

Right Ctrl

[13]:
```
TARGET     0
MESSAGE    0
dtype: int64
```

[14]:
```
df.duplicated().sum()
```

[14]:
```
np.int64(403)
```

[15]:
```
df.drop_duplicates(keep='first',inplace=True)
```

[16]:
```
df.duplicated().sum()
```

[16]:
```
np.int64(0)
```

[17]:
```
from sklearn.preprocessing import LabelEncoder
encoder=LabelEncoder()
df['TARGET']=encoder.fit_transform(df['TARGET'])
df['TARGET']
```

[17]:
```
0       0
1       0
2       1
3       0
4       0
       ..
5567    1
5568    0
5569    0
5570    0
5571    0
Name: TARGET, Length: 5169, dtype: int64
```

[18]:
```
df.head()
```

[18]:

	TARGET	MESSAGE
0	0	Go until jurong point, crazy.. Available only ...
1	0	Ok lar... Joking wif u oni...

Right Ctrl

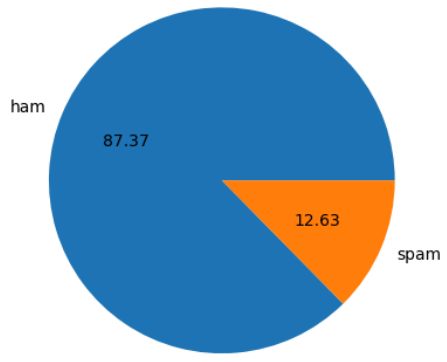
Name: TARGET, Length: 5169, dtype: int64

[18]:
```
df.head()
```

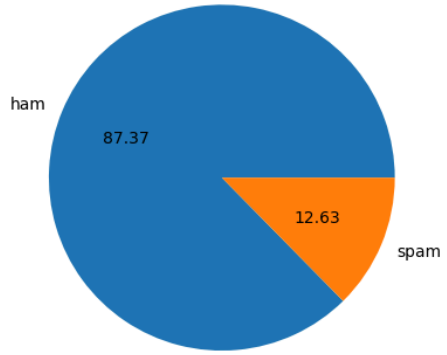
[18]:

	TARGET	MESSAGE
0	0	Go until jurong point, crazy.. Available only ...
1	0	Ok lar... Joking wif u oni...
2	1	Free entry in 2 a wkly comp to win FA Cup fina...
3	0	U dun say so early hor... U c already then say...
4	0	Nah I don't think he goes to usf, he lives aro...

[19]:
```
plt.pie(df["TARGET"].value_counts(),labels=['ham','spam'],autopct="%0.2f")
plt.show()
```



Right Ctrl



[20]:
```
x=df['MESSAGE']
y=df['TARGET']
```

[21]:
```
x
```

[21]:
```
0       Go until jurong point, crazy.. Available only ...
1                           Ok lar... Joking wif u oni...
2       Free entry in 2 a wkly comp to win FA Cup fina...
3       U dun say so early hor... U c already then say...
4       Nah I don't think he goes to usf, he lives aro...
                              ...
5567    This is the 2nd time we have tried 2 contact u...
5568                Will İ_ b going to esplanade fr home?
5569    Pity, * was in mood for that. So...any other s...
5570    The guy did some bitching but I acted like i'd...
5571                         Rofl. Its true to its name
```

Right Ctrl

[21]:
```
0       Go until jurong point, crazy.. Available only ...
1                           Ok lar... Joking wif u oni...
2       Free entry in 2 a wkly comp to win FA Cup fina...
3       U dun say so early hor... U c already then say...
4       Nah I don't think he goes to usf, he lives aro...
                              ...
5567    This is the 2nd time we have tried 2 contact u...
5568                Will İ_ b going to esplanade fr home?
5569    Pity, * was in mood for that. So...any other s...
5570    The guy did some bitching but I acted like i'd...
5571                         Rofl. Its true to its name
Name: MESSAGE, Length: 5169, dtype: object
```

[22]:
```
y
```

[22]:
```
0       0
1       0
2       1
3       0
4       0
       ..
5567    1
5568    0
5569    0
5570    0
5571    0
Name: TARGET, Length: 5169, dtype: int64
```

[23]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=3)
```

[24]:
```
print(x_train_cv)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[24], line 1
----> 1 print(x_train_cv)

NameError: name 'x_train_cv' is not defined
```

Right Ctrl

[25]:
```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn import svm
```

[26]:
```
cv=CountVectorizer()
```

[27]:
```
x_train_cv=cv.fit_transform(x_train)
x_test_cv=cv.fit_transform(x_test)
```

[28]:
```
print(x_train_cv)
```

```
<Compressed Sparse Row sparse matrix of dtype 'int64'
        with 54252 stored elements and shape (4135, 7641)>
  Coords        Values
  (0, 1879)     1
  (0, 1170)     1
  (0, 6840)     1
  (0, 6610)     1
  (0, 2779)     1
  (1, 1939)     1
  (1, 4467)     1
  (1, 453)      1
  (1, 7176)     1
  (1, 7594)     1
  (1, 1577)     1
  (1, 203)      1
  (1, 4768)     1
  (1, 7175)     1
  (1, 7390)     1
  (1, 7590)     1
  (1, 4309)     1
  (1, 5157)     1
  (1, 3732)     1
  (1, 3015)     1
  (1, 2333)     1
```

Right Ctrl

```
(1, 7594)     1
(1, 1577)     1
(1, 203)      1
(1, 4768)     1
(1, 7175)     1
(1, 7390)     1
(1, 7590)     1
(1, 4309)     1
(1, 5157)     1
(1, 3732)     1
(1, 3015)     1
(1, 2333)     1
(1, 5210)     1
(1, 4577)     1
(1, 4731)     1
(1, 5615)     1
:      :
(4134, 3290)  2
(4134, 4817)  1
(4134, 1546)  1
(4134, 4195)  1
(4134, 891)   1
(4134, 1092)  1
(4134, 1261)  1
(4134, 7302)  1
(4134, 6595)  1
(4134, 1624)  1
(4134, 1977)  1
(4134, 7438)  1
(4134, 6189)  1
(4134, 6815)  1
(4134, 2357)  1
(4134, 4093)  1
(4134, 6583)  1
(4134, 5934)  1
(4134, 1661)  1
(4134, 5153)  1
(4134, 6292)  1
(4134, 3707)  1
(4134, 6172)  1
```



[29]:
```
from sklearn.linear_model import LogisticRegression
lr=LogisticRegression()
```

[30]:
```
lr.fit(x_train_cv,y_train)
prediction_train=lr.predict(x_train_cv)
```

[31]:
```
from sklearn.metrics import accuracy_score
print(accuracy_score(y_train,prediction_train)*100)
```

```
99.75816203143893
```

[32]:
```
lr.fit(x_test_cv,y_test)
prediction_test=lr.predict(x_test_cv)
```

[33]:
```
from sklearn.metrics import accuracy_score
print(accuracy_score(y_test,prediction_test)*100)
```

```
99.80657640232108
```

Testing Result

Testing Email example

URGENT: Your account has been limited

Dear Customer,

We detected unusual activity on your account. For your safety, your account has been temporarily limited.

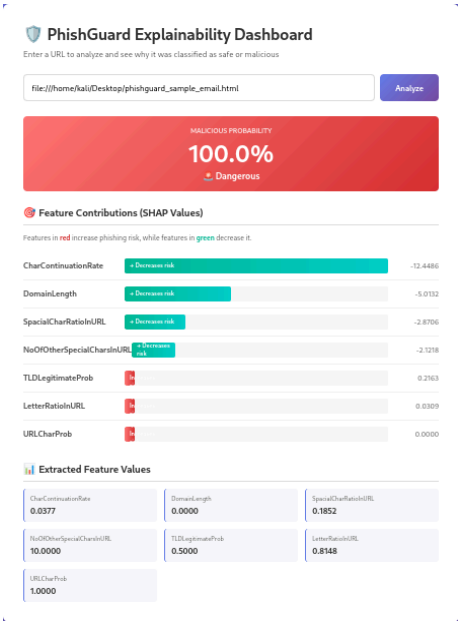
To avoid permanent suspension, you must verify your identity within 24 hours. Click the link below and enter your password and the One-Time Code sent to your phone.

https://secure.paypal-verify.example.org/account/login

Failure to verify will result in the closure of your account.

Sincerely,
PayPal Account Services

Phishing Guard AI

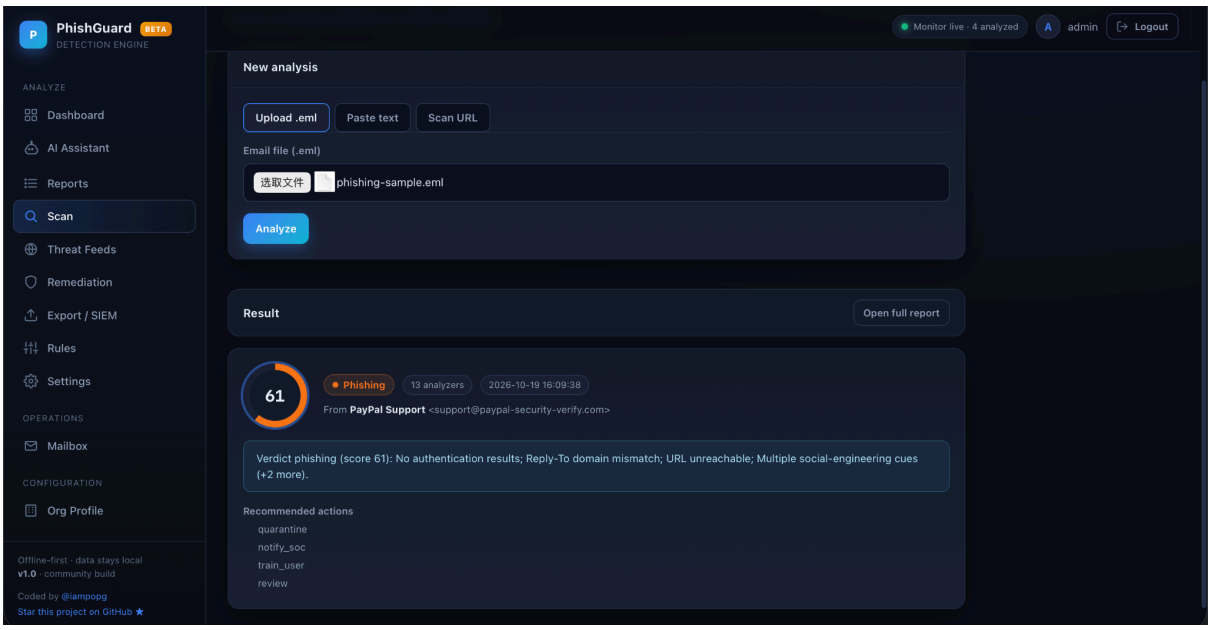


- cannot upload text, so I revise as html and upload
- file:///home/kali/Desktop/phishguard_sample_email.html

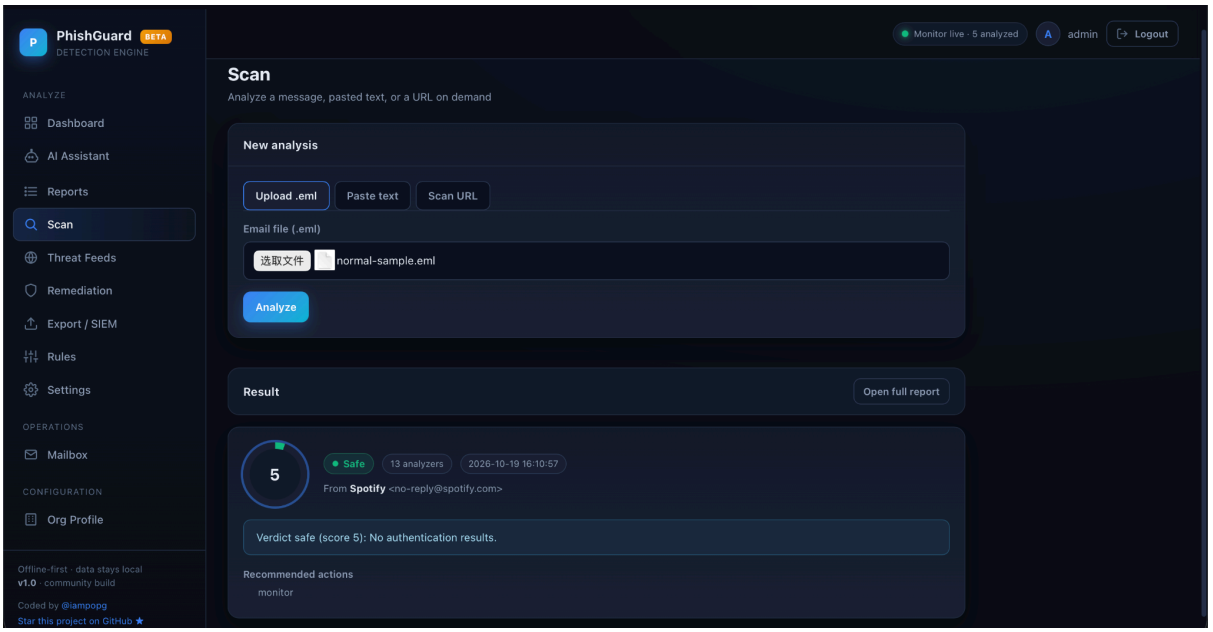
PhishGuard

Testing:

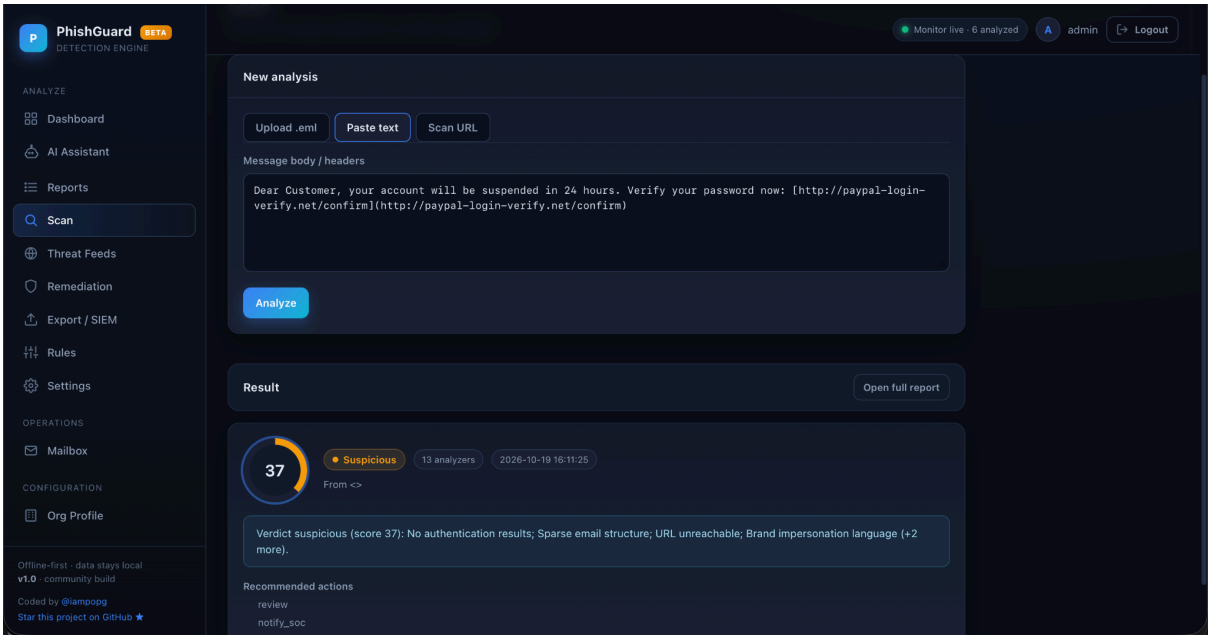
file_phishing-sample



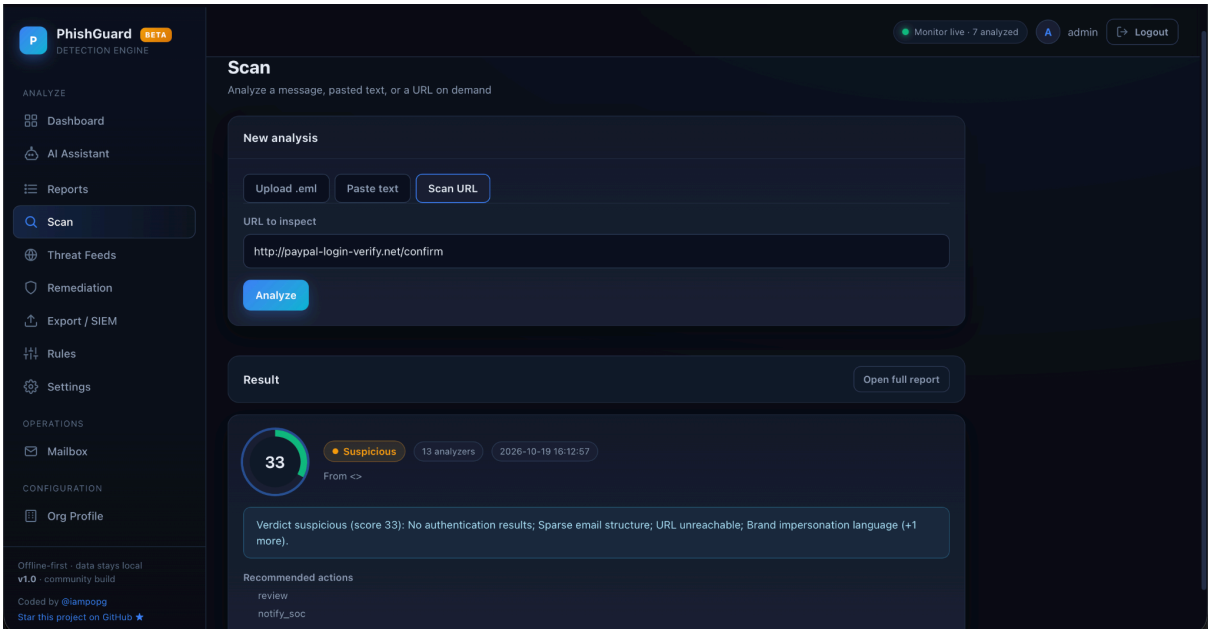
file_normal-sample:



Text:


PhishGuard
BETA
DETECTION ENGINE
ANALYZE
Dashboard
AI Assistant
Reports
Scan
Threat Feeds
Remediation
Export / SIEM
Rules
Settings
OPERATIONS
Mailbox
CONFIGURATION
Org Profile
Offline-first · data stays local
v1.0 · community build
Coded by @iampopg
Star this project on GitHub ★
Monitor live · 6 analyzed
A
admin
Logout
New analysis
Upload .eml
Paste text
Scan URL
Message body / headers
Dear Customer, your account will be suspended in 24 hours. Verify your password now: http://paypal-login-verify.net/confirm
Analyze
Result
Open full report
37
Suspicious
13 analyzers
2026-10-19 16:11:25
From <>
Verdict suspicious (score 37): No authentication results; Sparse email structure; URL unreachable; Brand impersonation language (+2 more).
Recommended actions
review
notify_soc

URL:


PhishGuard
BETA
DETECTION ENGINE
ANALYZE
Dashboard
AI Assistant
Reports
Scan
Threat Feeds
Remediation
Export / SIEM
Rules
Settings
OPERATIONS
Mailbox
CONFIGURATION
Org Profile
Offline-first · data stays local
v1.0 · community build
Coded by @iampopg
Star this project on GitHub ★
Monitor live · 7 analyzed
A
admin
Logout
Scan
Analyze a message, pasted text, or a URL on demand
New analysis
Upload .eml
Paste text
Scan URL
URL to inspect
http://paypal-login-verify.net/confirm
Analyze
Result
Open full report
33
Suspicious
13 analyzers
2026-10-19 16:12:57
From <>
Verdict suspicious (score 33): No authentication results; Sparse email structure; URL unreachable; Brand impersonation language (+1 more).
Recommended actions
review
notify_soc

NLP

[34]:
```python
import pandas as pd

# 1. 加载新测试数据（把 test_emails.csv 放到和 spam.csv 同一目录）
test_df = pd.read_csv("test_emails.csv", encoding='ISO-8859-1')

# 2. 预处理（和训练数据保持一致）
test_df = test_df.rename(columns={'v1': 'TARGET', 'v2': 'MESSAGE'})
test_df['TARGET'] = encoder.transform(test_df['TARGET'])  # 用之前已经fit好的encoder

# 3. 关键：用【已经fit好的 cv】transform 新数据，绝对不能再 fit！
X_new = test_df['MESSAGE']
X_new_cv = cv.transform(X_new)   # ← 注意是 transform，不是 fit_transform！

# 4. 用【已经训练好的 lr】预测，不是再 fit！
y_new = test_df['TARGET']
y_pred = lr.predict(X_new_cv)   # ← 注意是 predict，不是 fit！

# 5. 输出结果
print("=" * 60)
print(f"新测试集共 {len(test_df)} 条邮件")
print(f"准确率: {accuracy_score(y_new, y_pred) * 100:.2f}%")
print("=" * 60)
print()

for i, (msg, actual, pred) in enumerate(zip(X_new, y_new, y_pred)):
    label_actual = "spam" if actual == 1 else "ham"
    label_pred   = "spam" if pred == 1 else "ham"
    status = "✓" if actual == pred else "✗ 错判!"
    preview = msg[:60] + "..." if len(msg) > 60 else msg
    print(f"[{i+1:2d}] 真实={label_actual:4s}  预测={label_pred:4s}  {status}")
    print(f"      {preview}")
    print()
```

```
============================================================
新测试集共 30 条邮件
```

Right Ctrl

```
============================================================
新测试集共 30 条邮件
准确率: 93.33%
============================================================

[ 1] 真实=ham   预测=ham   ✓
      Hey, are we still grabbing lunch at noon tomorrow? Let me kn...

[ 2] 真实=ham   预测=ham   ✓
      The quarterly report draft is attached. Please review and le...

[ 3] 真实=ham   预测=ham   ✓
      Mom called, she wants to come visit next weekend. Are you fr...

[ 4] 真实=ham   预测=ham   ✓
      Can you forward me the meeting notes from yesterday's produc...

[ 5] 真实=ham   预测=ham   ✓
      Your Amazon package has been delivered to the front desk of ...

[ 6] 真实=ham   预测=ham   ✓
      Happy birthday! Hope you have an amazing day. Let's celebrat...

[ 7] 真实=ham   预测=ham   ✓
      The office wifi password has been updated. Please contact IT...

[ 8] 真实=ham   预测=ham   ✓
      Don't forget to submit your timesheet by the end of today. T...

[ 9] 真实=ham   预测=ham   ✓
      Hey, I'm running 10 minutes late for the 3pm meeting. Start ...

[10] 真实=ham   预测=ham   ✓
      The plumber called and said he can come tomorrow morning bet...

[11] 真实=ham   预测=ham   ✓
```

Right Ctrl

```
[11] 真实=ham   预测=ham   ✓
     Did you pick up the groceries on your way home? We need milk...

[12] 真实=ham   预测=ham   ✓
     Reminder: team building event this Friday at 5pm. Please RSV...

[13] 真实=ham   预测=ham   ✓
     Your dentist appointment is scheduled for October 22nd at 10...

[14] 真实=ham   预测=ham   ✓
     I put the key under the mat. Just let yourself in when you g...

[15] 真实=ham   预测=ham   ✓
     The weather forecast says it'll rain all day tomorrow, bring...

[16] 真实=ham   预测=ham   ✓
     Can you send me that spreadsheet we were working on? I need ...

[17] 真实=ham   预测=ham   ✓
     Dinner tonight at that new Italian place? I booked a table f...

[18] 真实=ham   预测=ham   ✓
     Your subscription renewal receipt is available in your accou...

[19] 真实=ham   预测=ham   ✓
     Please remember to lock the office door when you leave tonig...

[20] 真实=ham   预测=ham   ✓
     The cat vet appointment went well, she's perfectly healthy.

[21] 真实=spam  预测=spam  ✓
     URGENT: Your bank account has been temporarily suspended. Cl...

[22] 真实=spam  预测=spam  ✓
     CONGRATULATIONS! You have won a $1000 Amazon gift card! Clai...

[23] 真实=spam  预测=spam  ✓
     Dear PayPal customer, your account has been limited due to u...

[24] 真实=spam  预测=spam  ✓
```

```
     Your subscription renewal receipt is available in your accou...

[19] 真实=ham   预测=ham   ✓
     Please remember to lock the office door when you leave tonig...

[20] 真实=ham   预测=ham   ✓
     The cat vet appointment went well, she's perfectly healthy.

[21] 真实=spam  预测=spam  ✓
     URGENT: Your bank account has been temporarily suspended. Cl...

[22] 真实=spam  预测=spam  ✓
     CONGRATULATIONS! You have won a $1000 Amazon gift card! Clai...

[23] 真实=spam  预测=spam  ✓
     Dear PayPal customer, your account has been limited due to u...

[24] 真实=spam  预测=spam  ✓
     You have been selected to receive a FREE iPhone 15 Pro! Clic...

[25] 真实=spam  预测=ham   ✗ 错判!
     FINAL NOTICE: Your IRS tax payment is overdue in the amount ...

[26] 真实=spam  预测=spam  ✓
     WINNER! Your email has won Â£1,000,000 GBP in the Internatio...

[27] 真实=spam  预测=spam  ✓
     Dear Netflix member, your subscription has been put on hold....

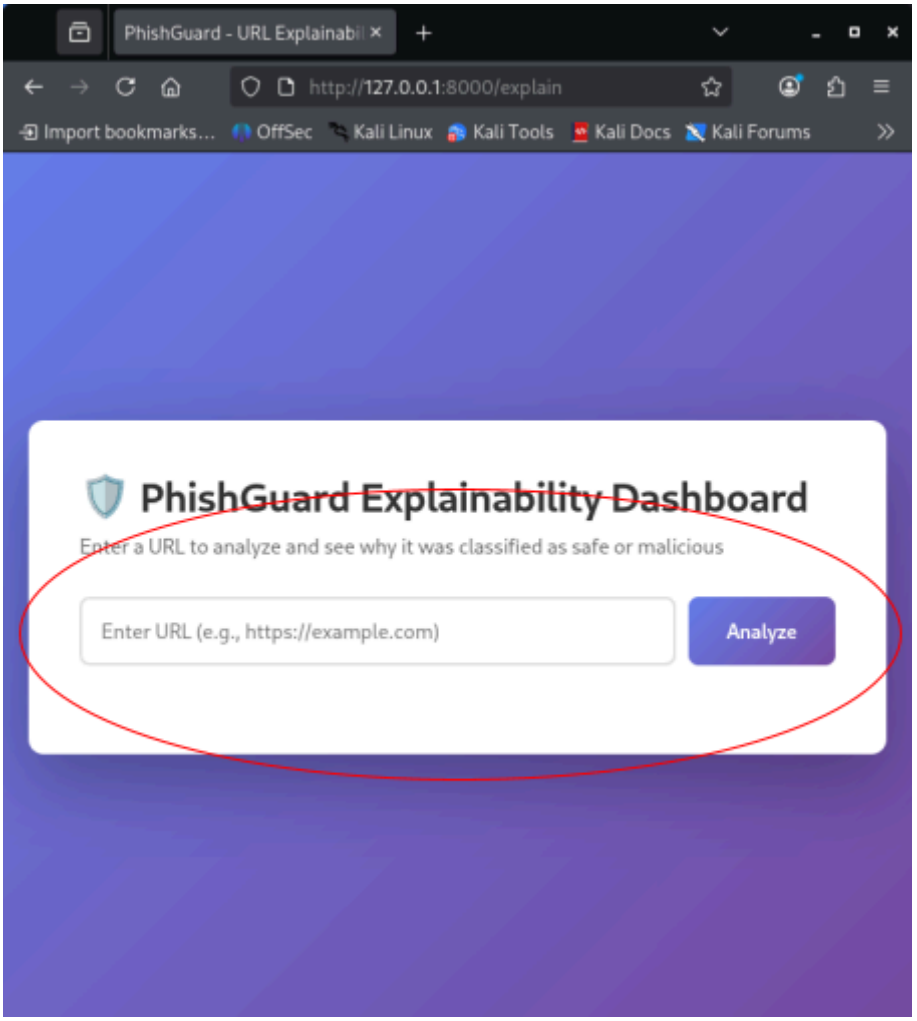
[28] 真实=spam  预测=spam  ✓
     ALERT: We detected a suspicious login on your Amazon account...

[29] 真实=spam  预测=spam  ✓
     FREE entry in our weekly competition to win a 7-day cruise t...

[30] 真实=spam  预测=ham   ✗ 错判!
     Dear Microsoft user, your email account will be permanently ...
```

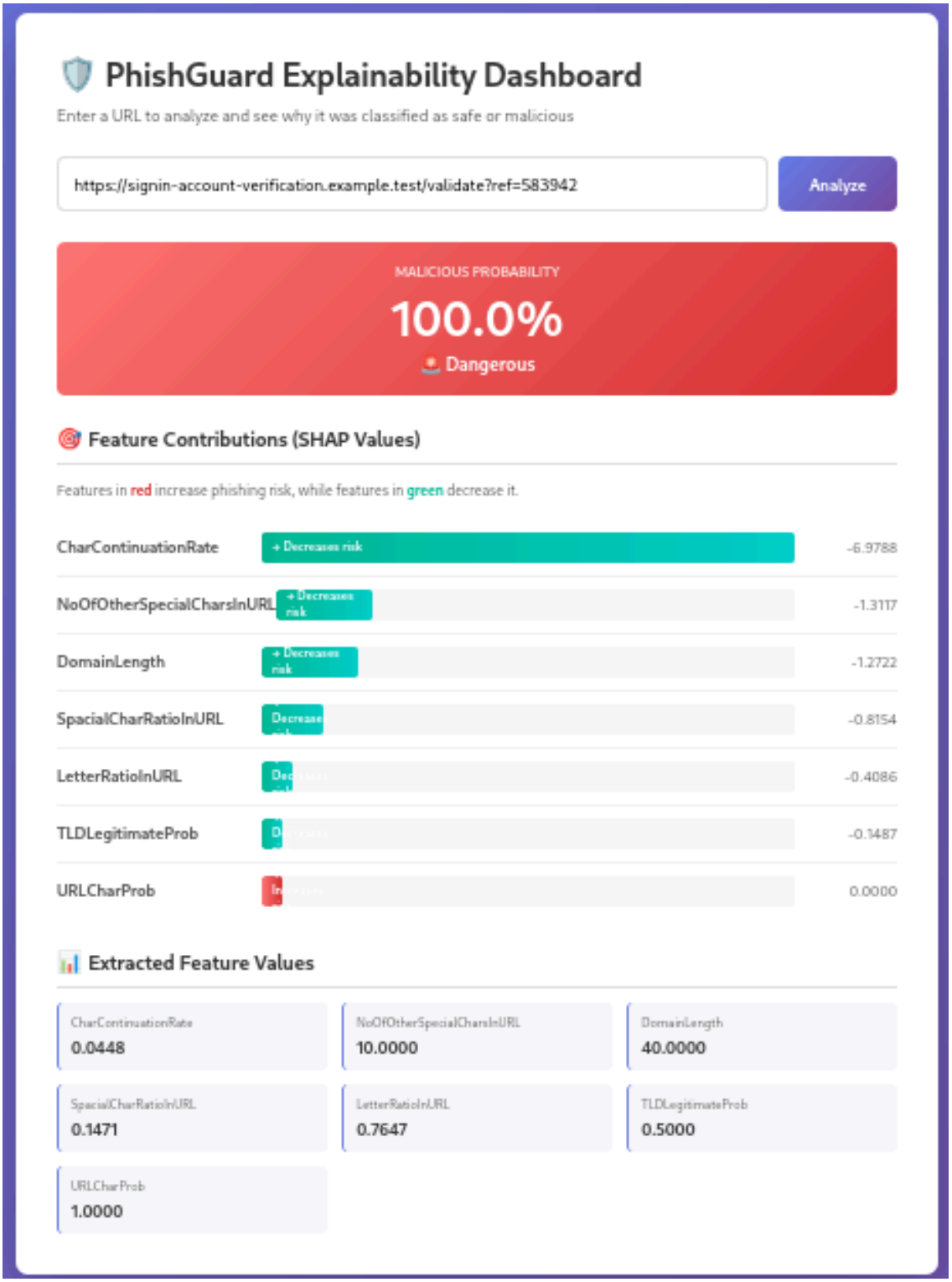
Summarize on the strengthens and weakness of those tools

Phishing Guard AI



Strengthens:

1. Interface : Easy to understand
2. ONLY place URL and click button to Analyze



Weakness:

1. Difficult to understand the metrics of each rule
2. What are the numbers actually means
3. Although the green bars is fulled, and less red bar, however still show 100% Malicious ; Hardly to understand how to calculate the result in interface

PhishGuard

Strengths

1. **Transparent.** Every score is traceable to a specific analyzer, with reasoning stated in each report.
2. **Accurate grading.** Your tests produced a clear gradient (61 / 5 / 37 / 33), and the normal email was not falsely flagged.
3. **Header analysis works.** The .eml version scored 24 points higher than the same text pasted in, thanks to header checks like Reply-To mismatch.
4. **Offline and private.** Runs fully locally with no data leaving your machine.
5. **Easy to use.** Upload, paste, or scan a URL — results appear in seconds.
6. **Free and open source.** MIT-licensed, no vendor lock-in.

Weaknesses

1. **Weak URL detection by default.** The lookalike domain wasn't flagged; "URL unreachable" added only 1 point in all four tests.
2. **Threshold is fragile.** The phishing sample scored 61, just 1 point above the line. Emails with valid auth headers could slip through.
3. **Most analyzers are dormant.** Sandbox, behavioral, and QR are off; threat-intel requires your own keys.
4. **Fixed 5-point penalty.** Any email without SPF/DKIM/DMARC headers gets dinged, even normal ones.
5. **Unfinished.** Remediation is a placeholder, single admin only, rules editable only in code.
6. **Heavy setup.** Needs Python + Node, slow first build, and it's missing python-multipart — the server crashed until we installed it.

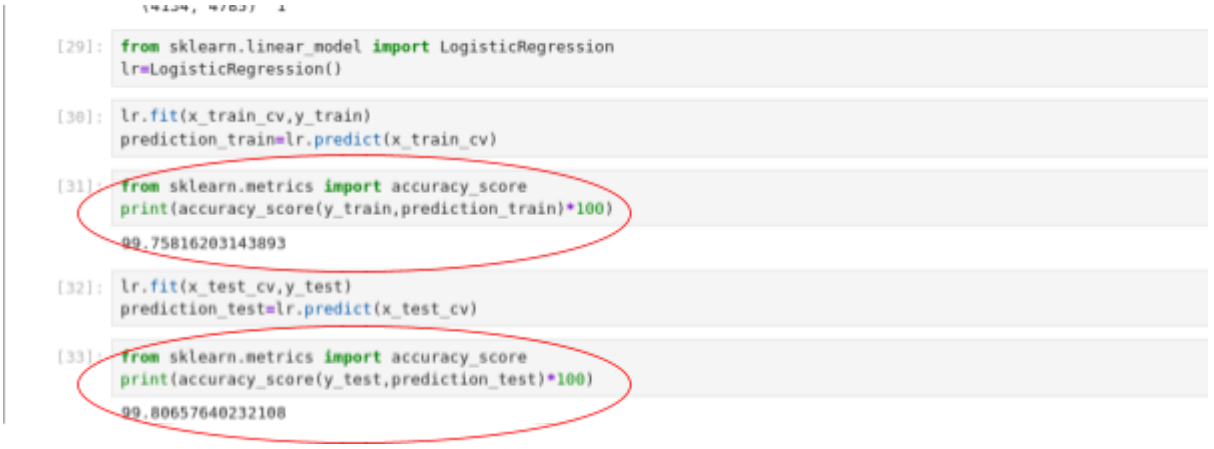
NLP

Strengthens:

1. Fast, easy to deploy and retrain.
2. Complete process

Weakness:

1. Difficult to understand



```
[29]: from sklearn.linear_model import LogisticRegression
      lr=LogisticRegression()

[30]: lr.fit(x_train_cv,y_train)
      prediction_train=lr.predict(x_train_cv)

[31]: from sklearn.metrics import accuracy_score
      print(accuracy_score(y_train,prediction_train)*100)

      99.75816203143893

[32]: lr.fit(x_test_cv,y_test)
      prediction_test=lr.predict(x_test_cv)

[33]: from sklearn.metrics import accuracy_score
      print(accuracy_score(y_test,prediction_test)*100)

      99.80657640232108
```

2. no semantic understanding , it misses meaning, context

Confirmed and signed by



2026-10-2026.